1. Clinical Question & Core Hypothesis

Clinical Question

Why do some MIBC tumors with robust local immune architecture (TLS/T-cell infiltration) nevertheless fail to benefit from immunotherapy? The Ontologist framed this as an interaction between a fibrogenic, TGF-β–dominated stroma and spatial immune organization (Immunoscore/TLS), with implications for immunotherapy response?

Working Hypothesis

Do TLS-like lymphoid aggregates in MIBC fail to confer benefit from immune checkpoint blockade when tumor TGFB1 expression is high, driving stromal fibrosis and functional exclusion of TLS leading to an ineffective Immunoscore? If true, TLS-rich/TGFB1-high tumors will exhibit reduced response to PD-1/PD-L1 inhibitors, indicating that combining TGF-β pathway inhibition with immunotherapy is required to restore TLS functionality and improve clinical outcomes.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	TGFB1, Immunoscore
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Immunoscore: Immunoscore/TLS path because the spatial organization and quality of lymphoid aggregates is the imaging/immune readout whose inconsistent prognostic value we need to explain; and (c) the immunotherapy-biomarker path because the clinical puzzle is specifically about therapeutic benefit from immune checkpoint blockade.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MSI	Supporting modifier from knowledge graph
Secondary	MSH6	Supporting modifier from knowledge graph

Secondary	pZVR	Supporting modifier from knowledge graph
Secondary	AI-based biomarkers	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective analysis of MIBC cohorts treated with PD-1/PD-L1 inhibitors to assess interactions between TGFB1 expression, markers of fibrosis (e.g., COL1A1, FN1), TLS presence/organization, Immunoscore, and objective response rate (ORR)/PFS.

Prediction 2: Spatial profiling (IHC, multiplex immunofluorescence, spatial transcriptomics) to quantify TLS maturation, T cell exclusion gradients, and ECM deposition in TGFB1-high vs TGFB1-low tumors.

Prediction 3: Preclinical modeling (mouse or organoid systems) to test whether TGFB1 overexpression drives fibrosis and TLS dysfunction and whether TGF-β blockade restores TLS function and enhances responsiveness to anti-PD-1/PD-L1.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MSI: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.